

Units

Explanations of Headings

Unit: The full name of a unit.

Abbreviation: Abbreviations per *Science* style (shown in boldface here only for emphasis).

Instructions for Handling:

sft: Spell out on first usage and use unit abbreviations as indicated thereafter.

dft: Define on first usage and use unit abbreviations as indicated thereafter. The editor should define in parentheses what the unit measures or what it is equivalent to, in most cases using the information provided in the Notes/Comments column.

A blank space means that the unit abbreviation should be used, without definition or explanation, throughout the text, references, tables, and figures. [In the abstract, all units (except °, °C, M, and %) are spelled out. See the Abstract section.]

Notes/Comments: General usage notes and definitions.

Système International (SI) Prefixes for Units

E	exa	10^{18}
P	peta	10^{15}
T	tera	10^{12}
G	giga	10^9
M	mega	10^6
k	kilo	10^3
m	milli	10^{-3}
μ	micro	10^{-6}
n	nano	10^{-9}
p	pico	10^{-12}

f	femto	10^{-15}
a	atto	10^{-18}
z	zepto	10^{-21}

SI Multiplier Prefixes

<u>Prefix</u>	<u>Symbol</u>	<u>Factor</u>
deka	da	10^1
hecto	h	10^2
kilo	k	10^3
mega	M	10^6
giga	G	10^9
tera	T	10^{12}
peta	P	10^{15}
exa	E	10^{18}
yotta	Y	10^{24}
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}
atto	a	10^{-18}
zepto	z	10^{-21}

(From *AIP Style Manual*, 1978, 3rd ed.)

<u>Unit</u>	<u>Abbreviation</u>	<u>Instructions for handling</u>	<u>Notes/Comments</u>
<u>A</u>			
<i>A</i>	absorbance (of light by solutions)	dft	example: absorbance at 260 nm (<i>A</i> ₂₆₀)
ampere	A		
ampere hour	A·hour		use center dot, not hyphen
angstrom	Å	okay to hyphenate when used as modifier (5-Å filament)	
arc minute	arc min		
arc second	arc sec	use "sec," not "s" (this is an exception to the use of "s" as abbreviation for second)	
astronomical unit	au	sft	
atmosphere	atm	this is not an SI unit but is allowed	
atomic mass unit	amu	outdated; query author if OK to change to the modern standard, u (unified atomic mass unit)	
atomic %	at %	sft	
atomic unit	au	sft	
attomole	amol	dft	1 amol = 10 ⁻¹⁸ mol
<u>B</u>			
bar		spell out	
barn	b	dft	unit of nuclear cross section equal to 10 ⁻²⁸ m ² per nucleus (derives from nuclear bombardment being as big as a barn)

barrels per day	bbl/day (as in oil)	sft	spell out unless used often; then use bbl/day (sft)
base pair	bp	sft; use abbreviation only with number	3- to 5-bp nucleotide (although bp is not a unit, treat it as one)
becquerel	Bq	sft	SI unit of radioactivity equal to one disintegration or other nuclear transformation per second; 1 curie = 37×10^9 Bq
<u>Unit</u>	<u>Abbreviation</u>	<u>Instructions for handling</u>	<u>Notes/Comments</u>
billion tonnes [or billion (metric) tons]	Gt	sft	don't use "gigatonne" unless no specific value is given (e.g., "at the gigatonne scale")
billion years	Gyr	sft	use only if repeated frequently [Commonly used for duration/length of time. Phrases may also be written out ("xx billion years").]
billion years (ago)	Ga	sft	use only if repeated frequently [Commonly used for age. "Ago" is often implied but may be written out ("xx billion years ago") when context is needed (follow authors).]
bohr		spell out	
bohr magneton	μ_B	sft	
British thermal unit	Btu	sft	plural: Btus
byte		spell out	
<u>C</u>			
calorie	cal		
Calorie	Cal		= 1000 calories or 1 kcal
candela	cd		

cells per milliliter	cells/ml		we allow cells/ml. Although a cell is not a unit, cells/ml comes up often in biological papers.
cells per well	cells per well	solidus is not allowed	
centimeter	cm		
centimorgan	cM	sft	
centipoise	cP	sft	a unit of viscosity equal to 1/100 poise
coulomb	C		
counts per minute	cpm	spell out only when used without a number: "... was measured in counts per minute."	
cubic centimeter	cm³	do not use cc	
curie	Ci	Ci/mmol = concentration Ci/ml = specific activity	
cycle		spell out	

<u>Unit</u>	<u>Abbreviation</u>	<u>Instructions for handling</u>	<u>Notes/Comments</u>
cycle per day	cycle/day		
cycle per degree	cycle/deg		
cycle per second		use Hz	

D

dalton	Da		unit of molecular mass equal to 1/12th the mass of a carbon-12 atom.
			kDa and MDa are ok.
day		spell out	

debye	D		
decibar	dbar		
decibel	dB	sometimes dBm is used for specific scale	
degrees (where referring to angle)		spell out unless preceded by a numeral, as in 13°	
degrees Celsius	°C		
degrees Fahrenheit	°F	convert to °C	
but: kelvin	K	kelvin does not use degree sign	formerly °K
disintegrations per minute	dpm		
Dobson unit	DU		1 DU = 10 ⁻³ cm (the thickness of the pure O ₃ layer that would be obtained if all the O ₃ in the vertical column were concentrated at normal temperature and pressure)
dyne		spell out	

<u>Unit</u>	<u>Abbreviation</u>	<u>Instructions for handling</u>	<u>Notes/Comments</u>
<u>E</u>			
einstein	E	dft	a unit that describes the photoenergy in a mole of a substance during a photochemical reaction (1 E = $Nh\nu$, where N is Avogadro's number, h is Planck's constant, and ν is

the frequency of the electromagnetic radiation). Define as follows: (energy of 1 mol of photons)

electromagnetic unit **emu**

sft

electron volt **eV**

electrostatic unit **esu**

sft

erg

spell out

equivalent **equiv**

do not add “s” to abbreviation

F

farad **F**

fathom

convert if possible

1 fathom = 6 feet

femtomole **fmol**

femtotesla **fT**

foot

convert to SI units

foot-candle

convert (10.7 lm/m²)

foot-lambert

convert (1.08 mL)

G

Galileo **gal**

dft

1 gal = 10⁻² m s⁻²

gallon

spell out (but usually convert)

gamma **γ**

γ = 10⁻⁵ oersted (a unit of magnetism) = 10⁻⁹ tesla

<u>Unit</u>	<u>Abbreviation</u>	<u>Instructions for handling</u>	<u>Notes/Comments</u>
γ (gamma)		change to mg if that is what is meant (in biology)	
gauss	G		1 G = 10^{-4} tesla
gibbs		spell out	unit of absorption
giga–electron volts	GeV		
gigahertz	GHz		
gigohm		spell out	
gilbert	Gb	dft	centimeter-gram-second (cgs); unit of magnetomotive force (not widely used)
Goldblatt unit	GU	sft	
gram	g		
gram atom		spell out	
gravity	<i>g</i>	italic and closed up; for example, 10,000 <i>g</i>	
gray	Gy	dft	SI unit of absorbed dose of ionizing radiation. Use instead of rads; 1 Gy = 100 rads
<u>H</u>			
heat flow units	HFU	sft	
hectare	ha		
henry	H	dft	meter-kilogram-second (mks); unit of self and mutual inductance
hertz	Hz		

horsepower **hp**

hour spell out

I

inch convert to SI

<u>Unit</u>	<u>Abbreviation</u>	<u>Instructions for handling</u>	<u>Notes/Comments</u>
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international units	IU		
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J

jansky	Jy	dft	unit of radio flux density (rate of flow of radio waves) (1 Jy = 10^{-26} W m ⁻² Hz ⁻¹)
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joule	J		
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joule electric	Je	sft	
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K

kayser		spell out	frequency of electromagnetic radiation (in cm ⁻¹)
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kelvin	K		write as: 200 K
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kilobar	kbar		
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kilobase	kb		
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kilo–base pair	kbp	sft	
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kilocalorie	kcal		
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kilocycles per second		use kHz	
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kilodalton	kDa		
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kilo–electron volt	keV		
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kilogauss	kG	
kilogram	kg	
kilogram-meter	kg·m	use center dot
kilohertz	kHz	
kilohm		spell out
kilojoule	kJ	
kilometer	km	
kilo-oersted	kOe	

<u>Unit</u>	<u>Abbreviation</u>	<u>Instructions for handling</u>	<u>Notes/Comments</u>
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kiloparsec	kpc	
kilopascal	kPa	
kilovolt	kV	
kilovolt peak	kVp	sft
kilowatt	kW	
kilowatt-hour	kWh	sft
kilowatt-hour electric	kWhe	sft
kilowatt thermal	kWt	sft
knots		spell out

L

lambert	L		
langley		spell out	unit of solar radiation density (1 langley = 697.8 W m ⁻² s ⁻¹)
Langmuir	usually L	dft	10 ⁻⁶ torr · s

light-year		spell out	unit of length
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liter		spell out when used alone but not in compound units such as ml
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lumen	lm	sft
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lux		spell out
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M

magnitude	<i>M</i>	dft
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body wave magnitude (earthquakes) (lc <i>m</i>)	<i>m_b</i>	dft
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moment magnitude (earthquakes)	<i>M_w</i>	dft
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surface wave magnitude (earthquakes)	<i>M_s</i>	dft
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mass of the Sun or solar mass	<i>M_☉</i>	dft
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maxwell	<i>Mx</i>
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megabase	<i>Mb</i>
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mega–base pairs	<i>Mbp</i>	sft
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megacycles per second		use MHz
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megadalton	<i>MDa</i>
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mega–electron volt	<i>MeV</i>
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megahertz	<i>MHz</i>
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megajoule **MJ**

<u>Unit</u>	<u>Abbreviation</u>	<u>Instructions for handling</u>	<u>Notes/Comments</u>
megavolts	MV	sft	
megawatt	MW		
megawatt-days (of fission energy released per kilogram of heavy metal)	MWd/kgHM	sft	
megawatt-electric	MWe	sft	
megawatt-hour	MWh	sft	
megawatt-thermal	MWt	sft	
megohm		spell out	
meter	m		
metric tons		spell out at first instance, use “tons” thereafter (or use “tonnes” - author preference)	
microampere	μA		
microcoulomb	μC		
micrometer	μm		
micromolar	μM		
micromole	μmol		
micron	μ	change to μm	
microsecond	μs		
mile		convert	
milliampere	mA		

milli–arc second	milli–arc sec
millibar	mbar
millicoulomb	mC
millicurie	mCi
milli–electron volt	meV
milliequivalent	mequiv
milligal	mgal

<u>Unit</u>	<u>Abbreviation</u>	<u>Instructions for handling</u>	<u>Notes/Comments</u>
milligram	mg		
millijoule	mJ		
milliliter	ml		
millimeters of mercury	mmHg		
millimicron	mμ	change to nm	
millimole	mmol		
million tonnes [or million (metric) tons]	Mt	sft	don't use “megatonne” unless no specific value is given (e.g., “at the megatonne scale”)
million years	Myr	sft	use only if repeated frequently [Commonly used for duration/length of time. Phrases may also be written out (“xx million years”).]
million years (ago)	Ma	sft	use only if repeated frequently [Commonly used for age. “Ago” is often implied but may be written out (“xx million years ago”) when context is needed (follow authors).]

milliosmol	mosmol	
minute	min	spell out only if no preceding numeral
molal	<i>m</i>	italicize em
molar	M	concentration. Example: 5 M or 72 mM
mole	mol	
mole %	mol %	
molecular weight		Generally spell out. Chemists have traditionally defined "molecular weight" as a unitless quantity (molecular weight ratio). Allow with or without units. However, polymer scientists commonly express molecular weight in units of g/mol.

<u>Unit</u>	<u>Abbreviation</u>	<u>Instructions for handling</u>	<u>Notes/Comments</u>
monolayer	ML	sft	
<u>N</u>			
nanosecond	ns		
nanometer	nm		
nanomole	nmol		
nautical mile		spell out	
newton	N		

normal	N	avoid	have author convert to molar (concentration)
nucleotide	nt	dft; use abbreviation only with a number	3- to 5-nt sequence (although nt is not a unit, treat it as one)

O

oersted	Oe		
ohm		spell out	do not use Ω
ohm-centimeter	ohm·cm	use center dot	
osmolal	osm		
osmolar	osM		
osmole	osmol		
ounce		convert	

P

parsec	pc	
parts per billion	ppb	sft
parts per million	ppm	sft
parts per thousand	ppt	sft

<u>Unit</u>	<u>Abbreviation</u>	<u>Instructions for handling</u>	<u>Notes/Comments</u>
parts per trillion by volume	pptv	sft	
pascal	Pa		
per mil	‰	sft	
picoampere	pA		

picomol	pmol		
poise	P	dft	1 P = 1 dyne·s cm ⁻²
pound		convert	
pounds per square inch	psi	convert if possible	

Q

quad	Q	Q = 10 ¹⁵ Btu	
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R

rad		(see gray)	
radian	rad	unit of angular measure	
rayleigh	R	sft	
Rayleigh number	Ra		dimensionless
reciprocal centimeter	cm⁻¹		
reciprocal ohm		use S for siemens	conductance 1 S = 1/ohm
research octane number	RON	sft	
revolutions per minute	rpm		
roentgen	R	sft	
roentgen-equivalent-man	rem	sft	
rydberg	Ry	sft	

<u>Unit</u>	<u>Abbreviation</u>	<u>Instructions for handling</u>	<u>Notes/Comments</u>
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S

second	s	use "sec" only in "arc sec"
siemens	S	1 S = 1 A/V (conductance unit)
steradian	sr	
svedberg	<i>S</i>	italic and closed up; for example, 5 <i>S</i> . Velocity of sedimentation of molecules per unit of accelerating field = the sedimentation coefficient. Values of the sedimentation coefficient lie in the range from 0.025 to 50×10^{-12} s. One svedberg = 10^{-13} s
sverdrup		spell out 1 sverdrup = a flow of ocean water of 10^6 cubic meters per second

T

tesla	T		1 nT = 1 gamma = 10^{-5} oersted; 1 T = 10^4 gauss
thousand years	kyr	sft	use only if repeated frequently [Commonly used for duration/length of time. Phrases may also be written out ("xx thousand years").]
thousand years (ago)	ka	sft	use only if repeated frequently [Commonly used for age. "Ago" is often implied but may be written out ("xx thousand years ago") when context is needed (follow authors).]
tonne			spell out; don't use "t" (or use "metric ton" - author preference)
torr		spell out	unit of pressure
troland		spell out	unit of retinal illuminance; retinal illumination produced

by a surface having a luminance of 1 candela per square meter, where the area of the aperture of the eye is 1 square millimeter.

U

unified atomic mass unit	u	sft	
unit	U		
universal time	UT	sft	(if used without a number; OK as is if used with a number)

<u>Unit</u>	<u>Abbreviation</u>	<u>Instructions for handling</u>	<u>Notes/Comments</u>
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V

volt	V		
volume %	vol %		v/v is usually preferred,; but be consistent within a paper
volume/volume (concentration)	v/v		
volume/weight	v/w		

W

watt	W		
weber	Wb	dft	SI unit of magnetic flux, equal to 10 ⁸ maxwells
weight hourly space velocity	WHSV	sft	
weight %	wt %		
weight/volume	w/v		

(concentration)

weight/weight

w/w

Y

year

spell out

years before the
present

yr B.P.

sft

[“calibrated years before the present (cal yr B.P.)”
is also permissible – follow authors]